

BASIC ELECTRONICS ENGINEERING

Paper-1123T

Semester-I

Time Allowed : 3 Hours] [Maximum Marks : 50

Note : The candidates are required to attempt **three** questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

SECTION—A

1. With a neat circuit diagram and waveforms explain the working of full wave bridge rectifier and show that its ripple factor is 0.48. 5

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2. Draw and explain the VI characteristics of UJT. 5
3. Design Zener voltage regulator for the following specifications : 5
- Input Voltage : $10V \pm 20\%$, Output Voltage = 5V,
 $I_L = 20 \text{ mA}$, $I_{Z\min} = 5 \text{ mA}$ and $I_{Z\max} = 80 \text{ mA}$.
4. Sketch the input and output characteristics of common emitter transistor configuration and explain briefly. 5
5. BJT can be used as an amplifier. Justify your answer. 5

SECTION—B

6. What you mean by Boolean algebra? Explain in detail the Demorgan's Theorem. 5
7. Perform the following operations : 5
- (a) $(111011101)_2 = (\quad)_{10}$
- (b) $(2F.C4)_{16} = (\quad)_8$

(d) $(374.26)_8 = (\quad)_{10}$

(e) $(82.25)_{10} = (\quad)_{16}$

8. Construct :

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(a) XOR gate using NAND gates only.

(b) XNOR gate using NOR gates only.

9. Draw and explain the block diagram of AM superheterodyne receiver system.

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10. What is the need of Modulation in a communication system? Explain.

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SECTION—C

11. Answer the following questions :

10×2=20

(i) Justify that $I_{CEO} \gg I_{CBO}$?

(ii) What is PIV ? What are its values for Half wave and Centre tapped Rectifiers?

(iii) What are Universal gates? Why are they called so?

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- (iv) Differentiate between different diode junction capacitances.
- (v) Draw the output waveform of capacitor filter with full wave rectifier.
- (vi) Give in tabular form the 4-bit binary code along with its equivalent gray code.
- (vii) Give symbol, logic expression and truth table of XNOR gate.
- (viii) Differentiate between modulation and

- (vii) Give symbol, logic expression and truth table of XNOR gate.
- (viii) Differentiate between modulation and demodulation.
- (ix) Give the frequency range of AM and FM transmission.
- (x) What values of A, B, C and D satisfy the following simultaneous Boolean equations?

$$\overline{A} + AB = 0, AB = AC, AB + A\overline{C} + CD = \overline{CD}.$$